

# A Fock-space inequality and the Laughlin spectral gap

OpenAI

September 24, 2026

## Abstract

We prove the spherical fermionic Laughlin spectral-gap conjecture for the full  $V_1$  interaction. With coefficient one for each pair projector, the gap above the Laughlin state at filling  $1/3$  on the round sphere is at least  $1/25$  for all sufficiently large systems. More generally,  $H_Q^2 \geq \gamma H_Q$  for some fixed  $\gamma > 1/25$  on the entire lowest-Landau-level Fock space for all sufficiently large flux  $Q$ , independently of particle number.

## 1 The gap and the model

Laughlin's wave function gave an explicit description of the fractional quantum Hall state at filling  $1/3$  [12]. Haldane's spherical geometry and pseudopotentials identify a particularly simple parent Hamiltonian: every pair pays energy one in relative angular momentum one and zero in the other pair sectors [7]. Its exact Laughlin zero mode is known, but a uniform lower bound on the positive spectrum requires control of all competing many-particle states. We prove this bound for the full interaction on the round sphere, resolving positively the spherical fermionic Laughlin spectral-gap conjecture.

The gap question has accompanied this model since its early development. Haldane and Rezayi's finite-size study supplied numerical evidence and explicitly distinguished the expected hard-core gap from a formal proof [8]. The density-wave variational picture of Girvin, MacDonald, and Platzman explains important features of the excitation spectrum [6]. A variational excitation energy is an upper bound; the lower bound sought here must hold for every state orthogonal to the zero modes.

### 1.1 Normalization and main results

For an integer  $Q \geq 1$ , let  $U_Q$  be the spin- $Q/2$  representation of  $SU(2)$ , equipped with its invariant inner product. It is the one-particle lowest Landau level on the sphere with  $Q$  flux quanta. Equivalently, it is the space of homogeneous polynomials of degree  $Q$  in spinor coordinates  $(u, v)$ . For  $2 \leq N \leq Q + 1$ , define

$$H_{N,Q} = \sum_{1 \leq i < j \leq N} P_{ij}^{(1)} \Big|_{\bigwedge^N U_Q}, \quad (1)$$

where  $P_{ij}^{(1)}$  is the orthogonal projector onto pair spin  $Q - 1$  in tensor positions  $i, j$ . This is relative angular momentum one. The sum preserves the antisymmetric subspace. Each unordered pair has coefficient one, with no rescaling depending on  $N$  or  $Q$ .

Set  $H_{0,Q} = H_{1,Q} = 0$ , and write

$$\mathcal{F}_Q = \bigoplus_{N=0}^{Q+1} \bigwedge^N U_Q, \quad H_Q = \bigoplus_{N=0}^{Q+1} H_{N,Q}, \quad \gamma_* = \frac{4616733319001}{10^{14}} > \frac{1}{25}.$$

Our principal estimate concerns this entire Fock space.

**Theorem 1.1.** *For every  $0 < \gamma < \gamma_*$ , there is an integer  $Q_\gamma$  such that for every integer  $Q \geq Q_\gamma$ ,*

$$H_Q^2 \geq \gamma H_Q \quad \text{on } \mathcal{F}_Q. \quad (2)$$

*The threshold  $Q_\gamma$  is independent of particle number.*

Thus the spectrum is contained in  $\{0\} \cup [\gamma, \infty)$ . In a particle sector without a zero mode, this statement bounds the energy away from zero; it does not assert a gap between that sector's two lowest eigenvalues. At Laughlin filling the zero mode is unique, and we obtain the usual ground-state spectral gap.

**Corollary 1.2.** *There is an integer  $N_0 \geq 2$  such that, for every  $N \geq N_0$  and  $Q = 3(N - 1)$ ,*

$$H_{N,Q} \geq \frac{1}{25}(I - P_{L,N}) \quad \text{on } \bigwedge^N U_Q, \quad (3)$$

*where  $P_{L,N}$  projects onto the line spanned by*

$$\Psi_{L,N} = \prod_{i < j} (u_i v_j - u_j v_i)^3.$$

At fixed magnetic length, the sphere radius is proportional to  $\sqrt{Q}$ , so this is the two-dimensional thermodynamic limit. The thresholds in Theorem 1.1 and Corollary 1.2 are existential. The proof does not give an explicit flux threshold or the endpoint coefficient  $\gamma_*$  on finite spheres.

## 1.2 Prior work

The exact ground-state structure is an established part of the theory. Besides Haldane's spherical construction, Trugman and Kivelson obtained exact Laughlin ground states for repulsive interactions in a vanishing-range limit [23]. The second-quantized algebra of pseudopotentials was developed systematically by Ortiz, Nussinov, Dukelsky, and Seidel [18]. Chen and Seidel, and Mazaheri, Ortiz, Nussinov, and Seidel, gave algebraic constructions and characterizations of their zero modes [2, 15]. These results identify the zero modes that a spectral estimate must preserve. We include a short self-contained proof of the known spherical zero-mode uniqueness at the Laughlin flux in Lemma 6.2.

For Laughlin states on a cylinder of any fixed radius, Jansen established thermodynamic correlation limits and axial periodicity [10]. Exact ground states of related finite-range orbital chains, including squeezed and matrix-product states, were constructed by Nakamura, Wang, and Bergholtz [17] and by Wang and Nakamura [24]. Nachtergaele, Warzel, and Young proved uniform gaps for a truncated fermionic pseudopotential [16]; Warzel and Young obtained thin-torus bounds that separate bulk behavior from open-chain edge modes [25]. Their invariant-subspace method also treats a truncated bosonic pseudopotential [26]. The fermionic estimates already hold uniformly across particle sectors. The distinction in Theorem 1.1 is the full spherical pair interaction, with its fixed projector normalization and sphere radius growing in the thermodynamic limit. The finite support used in our auxiliary comparison operators does not truncate that interaction.

Recent work gives further structural and spectral information. Lemm, Nachtergaele, Warzel, and Young derive recursions between particle sectors and compare charge and neutral gaps [13]; a uniform bound for the full pseudopotential still requires control of the model-specific ground-space overlap matrices in those criteria. Schraven and Warzel prove entanglement-gap and clustering results for Laughlin states on sufficiently thin cylinders [22]. The entanglement spectrum is distinct

from the Hamiltonian spectrum considered here. Density rigidity within the family of fully correlated Laughlin states is another complementary form of incompressibility, established by Lieb, Rougerie, and Yngvason [14]. No density-screening assumption enters our argument.

Rougerie’s survey records the spectral-gap conjecture and its geometric formulations [21, Appendix A]. For  $N$  fermions in the planar lowest Landau level, consider the sum over unordered pairs of the orthogonal projectors onto relative polynomial degree one, each with coefficient one and with center-of-mass degree unrestricted. Section 7 proves that its positive spectrum is bounded below by the endpoint  $\gamma_*$ , uniformly in particle number  $N \geq 2$  and total polynomial degree  $L$ . In particular, this proves the fermionic cubic case of the folklore spectral-gap conjecture in [21, Conjecture A.1], whose degree range is  $L \leq 3N(N-1)/2$ .

### 1.3 Proof strategy

Proving positivity of  $H^2 - \gamma H$  is a standard route to a gap, underlying finite-size criteria [11] and methods based on positive local decompositions [3, 20]. Our argument uses rotations and angular-momentum decomposition to construct such a comparison. It starts from the normal-ordered expansion of  $H_Q^2$ . It has two-, three-, and four-body terms. We compare them with the corresponding terms of a positive operator

$$\mathcal{K}_Q = (2Q - 1) \int_{SU(2)} \sum_{\rho=1}^7 F_\rho(g)^\dagger F_\rho(g) dg,$$

where each  $F_\rho$  is a specified linear combination of a pair annihilator and operators that annihilate three particles and create one. Here  $F_\rho(g)$  denotes rotation by  $g$ , and Haar measure has mass one. These auxiliary operators involve finitely many orbital modes; the Hamiltonian itself is never truncated.

Rotational symmetry reduces the comparison to fixed spin blocks. The three-body blocks have an explicit spectrum and a summable tail. For four particles, coupling the interacting pair to the different spin sectors of a second pair can produce several copies of the same total spin. Their four-particle images under exterior multiplication can be linearly dependent. For the retained blocks, seven rows of rational data establish finite inequalities in their large-flux limits. The remaining issue is to transfer those inequalities to finite spheres without an error that grows with the number of particles.

The key device is an exact diagonal change of variables for the pair coefficients. On a fixed set of orbital modes, it converts finite-flux annihilators into their limiting forms. Every perturbation is then bounded by the local pair energy. Rotation averaging turns this into  $o(1)H_Q$  on the full Fock space. This relative estimate remains valid when the Gram matrices of these limiting four-particle images have kernels. A second useful observation is to apply Bessel’s inequality to all equivalent spin copies at once; this substantially reduces the four-body comparison error.

The comparisons leave a positive margin:

$$H_Q^2 \geq \mathcal{K}_Q + (\gamma_* - o(1))H_Q. \tag{4}$$

All errors and convergence thresholds are independent of particle number. Figure 1 isolates the passage responsible for this uniformity. A small norm error on a growing many-particle space would not suffice: it must vanish on the zero modes and be controlled by the pair energy. The fixed-mode comparison gives precisely that control before rotations distribute it over the sphere.

Section 2 derives the normal-ordered square, and Section 3 establishes the spin formulas. Section 4 gives the finite comparisons. Section 5 proves their uniform transfer, and Section 6 controls the tail

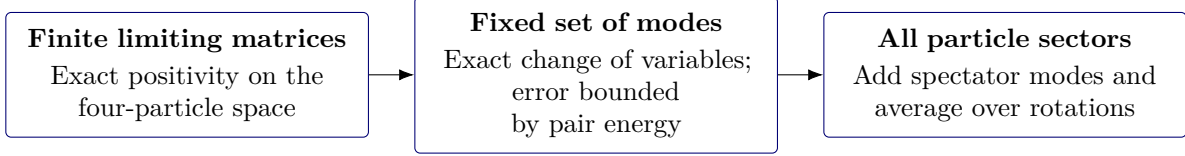


Figure 1: The finite-sphere comparison preserves an error relative to the pair energy at every step. The limiting positivity is imposed only after wedging; no invertibility of its Gram matrices is required.

and proves the spherical results. Section 7 derives the planar consequence. The exact row data, rational formulas, and complete arithmetic procedure for the positivity checks are in Appendix A.

## 2 Exterior algebra and the square of the interaction

Our first task is to isolate the two-, three-, and four-body terms of  $H_Q^2$ . The lifting notation below keeps this calculation independent of particle number.

Throughout the operator argument take  $Q \geq 2$ ; statements involving limits hold for all sufficiently large integer  $Q$ . We use the convention that wedges of distinct increasing orthonormal basis vectors have norm one. On

$$\mathcal{F}_Q = \bigoplus_{n=0}^{Q+1} \bigwedge^n U_Q,$$

let  $B(v)$  be the adjoint of left wedge multiplication by  $v$ . Thus  $B(v)$  is conjugate-linear in  $v$ , and

$$B(v \wedge w) = B(w)B(v).$$

For an orthonormal basis  $e_0, \dots, e_Q$  of  $U_Q$ , write  $c_j = B(e_j)$ . These operators satisfy the canonical anticommutation relations, and  $\|c_j\| \leq 1$ .

For a matrix on  $\bigwedge^k U_Q$ , define its lift by linear extension of

$$\mathcal{L}_k(|v\rangle\langle w|) = B(v)^\dagger B(w). \quad (5)$$

This map preserves positivity, intertwines rotations, acts as the original matrix on  $k$  particles, and vanishes on fewer than  $k$  particles. Positivity follows by decomposing a positive matrix into a sum of rank-one positive matrices.

Put  $a = Q - 1$  and let  $V_Q \subset \bigwedge^2 U_Q$  be the spin- $a$  summand. If  $v_0, \dots, v_{2Q-2}$  is an orthonormal basis of  $V_Q$ , set

$$B_p = B(v_p), \quad H_Q = \mathcal{L}_2(\Pi_{V_Q}) = \sum_{p=0}^{2Q-2} B_p^\dagger B_p. \quad (6)$$

This pair-annihilator form is standard in the algebraic treatment of pseudopotentials [18, Sections II.B–D]; see also [2, Equation (2)]. We verify its coefficient-one normalization here. Its restriction to  $\bigwedge^N U_Q$  is (1). Indeed contraction in two specified tensor positions corresponds to  $B(v)/\sqrt{\binom{N}{2}}$  under the isometric identification of wedge and antisymmetric tensor spaces; summing over the  $\binom{N}{2}$  unordered pairs gives (6).

Define intertwining wedge maps

$$W_3 : V_Q \otimes U_Q \longrightarrow \bigwedge^3 U_Q, \quad W_4 : V_Q \otimes V_Q \longrightarrow \bigwedge^4 U_Q,$$

by  $W_3(v \otimes u) = v \wedge u$  and  $W_4(v \otimes w) = v \wedge w$ . We suppress the subscript  $Q$  when no confusion can result.

A finite normal-ordering calculation in the sense of Wick [27] now determines the square. Only contractions between the two middle pair operators are needed.

**Lemma 2.1** (Normal-ordered square). *On  $\mathcal{F}_Q$ ,*

$$H^2 = H + \mathcal{L}_3(W_3(W_3^\dagger W_3 - I)W_3^\dagger) + \mathcal{L}_4(W_4 W_4^\dagger). \quad (7)$$

*Proof.* In normal ordering  $B_p^\dagger(B_p B_r^\dagger)B_r$ , only the two inner annihilators cross the two inner creators. Two, one, or zero contractions leave two-, three-, or four-body terms, respectively. On two particles,  $H = \Pi_{V_Q}$  is a projection, so the two-body coefficient is  $\Pi_{V_Q}$ . The term with no contractions is

$$\sum_{p,r} B(v_p \wedge v_r)^\dagger B(v_p \wedge v_r) = \mathcal{L}_4(W_4 W_4^\dagger).$$

There is no extra factor of two: this sum, and the defining basis of  $V_Q \otimes V_Q$ , both use ordered pairs. On three particles,  $H = W_3 W_3^\dagger$ , since  $W_3$  restricted to  $v_p \otimes U_Q$  is the map  $B_p^\dagger$  from one to three particles. Subtracting its two-body contribution from  $H^2$  therefore determines the remaining coefficient as

$$(W_3 W_3^\dagger)^2 - W_3 W_3^\dagger = W_3(W_3^\dagger W_3 - I)W_3^\dagger.$$

Normal-ordered coefficients are determined successively by these restrictions, proving the identity on Fock space.  $\square$

### 3 Spin coefficients and their limits

The three-body comparison will use the exact spin spectrum of the wedge map. The four-body comparison will use only finitely many coupling coefficients and their limits.

The spherical pair coefficients can be expressed through angular-momentum coupling [18, Section II.C and Appendix A]. We derive the particular coefficients and fixed-deficit limits needed here to keep their phases and normalization explicit.

We now fix  $e_p$  and  $v_p$  to be the normalized lowering bases of  $U_Q$  and  $V_Q$ , respectively, with  $v_0 = -e_0 \wedge e_1$ . For spin  $j$ , lowering from label  $p$  to  $p+1$  has coefficient  $\sqrt{(p+1)(2j-p)}$ . Write  $(s)_p = s(s-1)\cdots(s-p+1)$ , with  $(s)_0 = 1$ .

**Lemma 3.1** (Pair coefficients). *The coefficient of  $e_i \wedge e_j$  in  $v_p$ , for  $i < j$ , vanishes unless  $i+j = p+1$ , and otherwise equals*

$$v_p(i, j) = (i-j) \sqrt{\frac{(Q)_i (Q)_j p!}{Q(2Q-2)_p i! j!}}. \quad (8)$$

*For fixed  $p, i, j$ , its limit is*

$$v_p^*(i, j) = (i-j) \sqrt{\frac{p!}{2^p i! j!}} \mathbf{1}_{i+j=p+1}. \quad (9)$$

*Proof.* Apply simultaneous lowering  $p$  times to  $-e_0 \wedge e_1$  and divide by  $\sqrt{(2Q-2)_p p!}$ . Expand the lowering operator binomially. Combining the two contributions to each increasing pair gives (8). Taking the limit gives (9). The normalization can also be checked directly: the square of the coefficient in (8) is

$$(i-j)^2 \frac{\binom{Q}{i} \binom{Q}{j}}{Q \binom{2Q-2}{p}},$$

and differentiating  $(1+x)^Q$  gives

$$\sum_{i,j=0}^Q (i-j)^2 \binom{Q}{i} \binom{Q}{j} x^{i+j} = 2Qx(1+x)^{2Q-2}.$$

Take the coefficient of  $x^{p+1}$  and halve the sum to restrict to  $i < j$ .  $\square$

The elementary tensor product rule decomposes spins  $j_1, j_2$  into one copy of each spin  $j_1 + j_2 - z$ , for  $0 \leq z \leq \min(2j_1, 2j_2)$ . This rule can be seen directly in the polynomial model: at deficit  $z$  the raising equation determines a highest vector, and normalized lowering generates its  $2(j_1 + j_2 - z) + 1$  orthogonal weight vectors. Different highest spins give orthogonal invariant subspaces, and the sum of these dimensions is  $(2j_1 + 1)(2j_2 + 1)$ , exhausting the tensor product. Let  $R_{3,z}$  be the corresponding projection in  $V_Q \otimes U_Q$ , and put

$$d_a = 2Q - 1, \quad d_{3,z} = 3Q - 1 - 2z.$$

For  $Q \geq 2$ , the allowed indices are  $0 \leq z \leq Q$ .

**Lemma 3.2** (Three-body Gram eigenvalues).

$$W_3^\dagger W_3 - I = \sum_{z=0}^Q q_z(Q) R_{3,z}, \quad q_z(Q) = (-1)^z \frac{(Q)_z}{(2Q-2)_z} \left( 3z - 1 - \frac{z(z+1)}{Q} \right). \quad (10)$$

In particular  $W_3 R_{3,z} = 0$  for  $z = 0, 1, 3$  whenever these indices are allowed. For fixed  $z$ ,

$$q_z(Q) \longrightarrow (3z - 1)(-1/2)^z. \quad (11)$$

*Proof.* Identify  $V_Q \otimes U_Q$  with tensors antisymmetric in their first two positions. Then  $W_3$  is  $\sqrt{3}$  times full antisymmetrization. Its Gram operator is the compression of  $I - P_{13} - P_{23}$ , where  $P_{ij}$  swaps tensor positions. The two swap compressions coincide, since conjugation by  $P_{12}$  interchanges them and  $P_{12} = -I$  on the domain. By equivariance and multiplicity-freeness they act by the same scalar  $b_z$  on each spin block.

A highest vector of that block has coefficients  $s_p$  on  $v_p \otimes e_{z-p}$ ,  $0 \leq p \leq z$ , satisfying

$$\frac{s_{p+1}}{s_p} = -\sqrt{\frac{(z-p)(Q-z+p+1)}{(p+1)(2Q-2-p)}}. \quad (12)$$

This follows by applying the raising operator. Write  $\xi_z = \sum_{p=0}^z s_p v_p \otimes e_{z-p}$ . Since  $v_0 = (-e_0 \otimes e_1 + e_1 \otimes e_0)/\sqrt{2}$ , testing  $P_{23}\xi_z$  against  $v_0 \otimes e_z$  tests  $\xi_z$  only on the ordered triples  $(0, z, 1)$  and  $(1, z, 0)$ . Their pair labels are  $z-1$  and  $z$ , respectively, with the first contribution absent at  $z=0$ . Substituting their pair coefficients from (8) gives

$$s_0 b_z = \frac{1}{2} \left[ s_{z-1} \sqrt{\frac{z(Q)_z}{Q(2Q-2)_{z-1}}} - s_z (z-1) \sqrt{\frac{(Q)_z}{(2Q-2)_z}} \right].$$

Here the ordered tensor coefficients of a normalized wedge include a factor  $1/\sqrt{2}$ . Iteration of (12) gives

$$\frac{s_z}{s_0} = (-1)^z \sqrt{\frac{(Q)_z}{(2Q-2)_z}}, \quad \frac{s_{z-1}}{s_0} = (-1)^{z-1} \sqrt{\frac{z(Q-1)_{z-1}}{(2Q-2)_{z-1}}} \quad (z \geq 1).$$

Substitution gives  $q_z = -2b_z$  in the form (10). The three stated null sectors have  $q_z = -1$ , and (11) follows directly.  $\square$

We will repeatedly use a fixed-deficit limit of this same calculation. In a product of spins  $j_1, j_2$ , total lowering deficit means  $T = j_1 + j_2 - m$ , where  $m$  is the total magnetic quantum number. Coefficients below are with respect to normalized monomials  $X^p Y^{T-p} / \sqrt{p!(T-p)!}$ , and are zero outside the stated index ranges.

**Lemma 3.3** (Fixed-deficit coupling limit). *Suppose  $j_1, j_2 \rightarrow \infty$  and  $j_2/(j_1 + j_2) \rightarrow u^2$ , where  $0 < u < 1$ . Put  $v = \sqrt{1 - u^2}$ . For fixed integers  $0 \leq z \leq T$ , the normalized coupled vector of spin  $j_1 + j_2 - z$  at total lowering deficit  $T$  can be chosen with real coefficients converging to the coefficients  $s_{z,T,p}^{(u)}$  of*

$$\frac{(uX - vY)^z (vX + uY)^{T-z}}{\sqrt{z!(T-z)!}}. \quad (13)$$

*Descendants in every copy are obtained by normalized lowering, so the matching of bases between equal-spin copies is intertwining.*

*Proof.* At  $T = z$ , the raising equation gives the adjacent ratio

$$-\sqrt{\frac{(z-p)(2j_2 - z + p + 1)}{(p+1)(2j_1 - p)}}.$$

With sign  $(-1)^z$  at  $p = 0$ , normalization yields the coefficients of  $(uX - vY)^z / \sqrt{z!}$  in the limit. The change of variables  $(X, Y) \mapsto (uX - vY, vX + uY)$  is orthogonal, so these polynomials have norm one in the normalized-monomial inner product. Normalized lowering from  $T$  to  $T + 1$  divides the sum of the two lowering operators by

$$\sqrt{(T - z + 1)\{2(j_1 + j_2 - z) - (T - z)\}}.$$

In the limit this acts by multiplication by  $(vX + uY) / \sqrt{T - z + 1}$ . Induction proves the statement for fixed descendants.  $\square$

## 4 Rotational averages of explicit squares

We construct a positive rotational average and compare its three- and four-body terms with those of  $H_Q^2$ . This section specifies the finite inequalities needed; Appendix A proves them by exact rational arithmetic.

Haar measure  $dg$  on  $SU(2)$  is normalized to have total mass one. For an operator  $A$ , write  $A(g) = U(g)AU(g)^\dagger$ , with the appropriate Fock representation. Schur averaging sends a rank-one operator on an irreducible spin- $j$  space to its trace times  $I/(2j + 1)$ . For several equivalent copies, it acts in the same way on the spin factor and retains the matrix between copies. Explicitly, if the representation is  $\bigoplus_j (V_j \otimes \mathbb{C}^{m_j})$  and  $P_j$  is its spin- $j$  projection, then

$$\int U(g)XU(g)^\dagger dg = \bigoplus_j \frac{I_{V_j}}{2j + 1} \otimes \text{Tr}_{V_j}(P_j X P_j).$$

Here  $\text{Tr}_{V_j}$  is the partial trace over the spin factor. This follows from Schur's lemma applied to each pair of irreducible copies, with the scalar fixed by its trace; see [1, Section II.C]. In particular, averaging preserves off-diagonal entries between equivalent spin copies.

The positive-square construction belongs to the quadratic-gap approach described in Section 1. Fermionic positivity constraints, normal ordering, and symmetry also appear in the quantum Hall

bootstrap of Gao et al. [4], which bounds ground-state energies. Here the explicit squares are compared with  $H_Q^2 - \gamma H_Q$  to control the positive excitation spectrum.

For each of the seven rows in Appendix A, the integer  $t_\rho$  lies in  $\{0, \dots, 7\}$ . Set

$$\mathcal{S}_\rho = \{(p, j) \in \mathbb{Z}^2 : 0 \leq p \leq 7, 0 \leq j \leq 8, 0 \leq p + j - t_\rho \leq 8\}.$$

The row specifies real coefficients  $\lambda_\rho$  and  $\alpha_{pj}^\rho$  for  $(p, j) \in \mathcal{S}_\rho$ ; unlisted entries are zero. Every coefficient sum for row  $\rho$  below is over  $\mathcal{S}_\rho$ . Our comparison operator is

$$K = \sum_{\rho=1}^7 F_\rho^\dagger F_\rho, \quad F_\rho = \lambda_\rho B_{t_\rho} + \sum_{(p,j) \in \mathcal{S}_\rho} \alpha_{pj}^\rho c_{p+j-t_\rho}^\dagger c_j B_p. \quad (14)$$

Each summand of  $F_\rho$  removes two particles and decreases the sum of occupied orbital labels by  $t_\rho + 1$ : for the second term the decrease is  $(p+1) + j - (p+j-t_\rho) = t_\rho + 1$ . This common weight allows the two parts to interfere in their square, adjusting the three-body comparison at the two-body cost  $\lambda_\rho^2 B_{t_\rho}^\dagger B_{t_\rho}$ . Only modes  $0, \dots, 8$  occur in these auxiliary squares.

For a single row, omit  $\rho$  and put  $i = p + j - t$ ,  $k = q + l - t$ . Define

$$O_{p,j,k} = B(v_p \wedge e_j \wedge e_k) = c_k c_j B_p.$$

Reordering  $c_i c_k^\dagger = \delta_{ik} - c_k^\dagger c_i$  gives the two-, three-, and four-body terms of  $F^\dagger F$ :

$$\lambda^2 B_t^\dagger B_t, \quad (15)$$

$$\sum_{p,j} \lambda \alpha_{pj} \left[ (c_i B_t)^\dagger c_j B_p + (c_j B_p)^\dagger c_i B_t \right] + \sum_{p,j;q,l} \alpha_{pj} \alpha_{ql} \delta_{ik} (c_j B_p)^\dagger c_l B_q, \quad (16)$$

$$- \sum_{p,j;q,l} \alpha_{pj} \alpha_{ql} O_{p,j,k}^\dagger O_{q,l,i}. \quad (17)$$

Let  $A_2, A_3, A_4$  denote these terms, summed over rows and averaged with factor  $d_a$ . Thus

$$\mathcal{K}_Q := d_a \int K(g) dg = A_2 + A_3 + A_4, \quad A_2 = \eta H, \quad \eta = \sum_\rho \lambda_\rho^2. \quad (18)$$

#### 4.1 The three-body comparison

Before wedging, (16) is a matrix on  $V_Q \otimes U_Q$ . Its nonzero entries preserve  $T = p + j$ , and  $T \leq 15$ . Averaging therefore gives

$$A_3 = \mathcal{L}_3 \left( W_3 \sum_{z=0}^{15} \frac{d_a}{d_{3,z}} e_z^Q R_{3,z} W_3^\dagger \right), \quad (19)$$

for all sufficiently large  $Q$ , where  $e_z^Q$  is the trace of the unaveraged matrix in that spin block. By Lemma 3.3, with  $u = 1/\sqrt{3}$ , these traces converge to

$$e_z = \sum_\rho \sum_{T=\max(z, t_\rho)}^{15} \left[ 2\lambda_\rho s_{z,T,t_\rho} \sum_{p+j=T} \alpha_{pj}^\rho s_{z,T,p} + \left( \sum_{p+j=T} \alpha_{pj}^\rho s_{z,T,p} \right)^2 \right], \quad (20)$$

where  $s = s^{(1/\sqrt{3})}$  and empty inner sums are zero. The condition  $i = k$  in (16) is exactly the equality of the two values of  $T$ .

The exact calculation in Appendix A proves

$$e_z < \frac{3}{2}(3z-1)(-1/2)^z, \quad z \in \{2, 4, 5, \dots, 15\}. \quad (21)$$

There are finitely many strict inequalities,  $d_{3,z}/d_a \rightarrow 3/2$ , and the omitted blocks  $z = 0, 1, 3$  are annihilated by  $W_3$ . Consequently, for all sufficiently large  $Q$ ,

$$A_3 \leq \mathcal{L}_3 \left( W_3 \sum_{z=0}^{15} q_z(Q) R_{3,z} W_3^\dagger \right). \quad (22)$$

## 4.2 The four-body comparison before taking limits

Let

$$\widehat{W} : V_Q \otimes \bigwedge^2 U_Q \longrightarrow \bigwedge^4 U_Q$$

be the wedge map. The second tensor factor decomposes into spins  $Q - r$ , with  $r$  odd,  $1 \leq r \leq Q$ . In its lowering bases use the sign of Lemma 3.3; at  $r = 1$  this is precisely the basis  $v_p$ .

A copy of spin  $2Q - 1 - D$  is obtained by coupling  $a = Q - 1$  with  $Q - r$  at deficit  $D - r$ . Denote its vector at total deficit  $T$  by  $|D, T; r\rangle_Q$ . Its physical orbital-index sum is  $1 + T$ , and it is a highest vector when  $T = D$ . For fixed  $D, T$  and sufficiently large  $Q$ , the allowed copy labels are

$$\mathcal{R}_D = \{1, 3, 5, \dots\} \cap [1, D], \quad D \leq T.$$

Define the real finite-flux coefficients by

$$|D, T; r\rangle_Q = \sum_{\substack{p+j+k=T \\ j < k}} S_r^Q(D, T; p, j, k) v_p \otimes (e_j \wedge e_k),$$

and extend them antisymmetrically in  $j, k$ . By two applications of Lemma 3.3, they converge to

$$S_r(D, T; p, j, k) = \sqrt{2} s_{r,j+k,j}^{(1/\sqrt{2})} s_{D-r,T-r,p}^{(1/\sqrt{2})}, \quad (23)$$

interpreted as zero if  $r > j + k$ . The formula holds for either ordering of  $j, k$ , by antisymmetry. It follows by applying Lemma 3.3 first to the two single-particle spins and then to the two pair spins.

Only  $1 \leq D \leq 23$  contribute: every unaveraged term has

$$T = p + j + k = q + l + i \leq 23.$$

Put  $d_D = 4Q - 1 - 2D$ . In the coupled bases  $|D, T; r\rangle_Q$ , the spin- $2Q - 1 - D$  block is identified with  $\mathbb{C}^{d_D} \otimes \mathbb{C}^{\mathcal{R}_D}$ : the first factor uses the  $(T - D)$ -th normalized spin descendant, and the second has basis vector indexed by  $r$ . Let  $E_D^Q$  be the partial trace over the spin factor of the compression to this block of the matrix representing (17) before wedging, summed over rows. Schur averaging then gives, for all sufficiently large  $Q$ ,

$$A_4 = \mathcal{L}_4 \left( \widehat{W} \left[ \bigoplus_{D=1}^{23} \frac{d_a}{d_D} (I_{d_D} \otimes E_D^Q) \right] \widehat{W}^\dagger \right), \quad (24)$$

where the direct sum acts as zero on all other spin blocks. For these finitely many  $D$ , entrywise convergence gives  $E_D^Q \rightarrow E_D$ , with

$$(E_D)_{rs} = - \sum_{\rho} \sum_{\substack{p,j,q,l \\ T \geq D}} \alpha_{pj}^{\rho} \alpha_{ql}^{\rho} S_r(D, T; p, j, k) S_s(D, T; q, l, i), \quad (25)$$

where  $i = p + j - t_\rho$ ,  $k = q + l - t_\rho$ , and  $T = p + j + k$ . The exact finite-flux matrix  $E_D^Q$  is given by the same formula with each  $S_r$  replaced by  $S_r^Q$ . Exchanging the two ordered entry pairs exchanges  $r$  and  $s$ , so both matrices are real symmetric.

The target term  $\mathcal{L}_4(W_4 W_4^\dagger)$  is obtained from the  $r = 1$  summand, since that summand is  $V_Q \otimes V_Q$ . Thus its copy matrix, in the units just used, is  $(d_D/d_a)\chi$ , where

$$\chi_{rs} = \mathbf{1}_{r=s=1}, \quad d_D/d_a \rightarrow 2.$$

For even  $D$ , the  $r = 1$  coupled vector is antisymmetric under exchange of its two spin- $a$  factors, since its coupling deficit is  $D - 1$ . Wedging two degree-two vectors is symmetric, so this target column then vanishes after applying  $\widehat{W}$ . The formula with  $\chi$  still represents it exactly; we never assume that the wedged copy vectors are independent.

Target spin blocks outside  $1 \leq D \leq 23$  contribute positively and may be discarded in a lower bound.

Set

$$w_{Dr}^Q = \widehat{W}|D, D; r\rangle_Q, \quad (26)$$

$$w_{Dr}^* = \sum_{\substack{p+j+k=D \\ j < k}} S_r(D, D; p, j, k) v_p^* \wedge e_j \wedge e_k, \quad (G_D)_{rs} = \langle w_{Dr}^*, w_{Ds}^* \rangle. \quad (27)$$

The star is a limit label, not an adjoint. Let  $W_D^*$  be the map with columns  $w_{Dr}^*$ . With allowance  $\varepsilon = 3 \cdot 10^{-6}$ , the limiting comparison on these four-particle images asks that  $W_D^*(2\chi - E_D + \varepsilon I)(W_D^*)^\dagger$  be nonnegative. Lemma 5.1 identifies this with the exact certificate

$$G_D(2\chi - E_D + \varepsilon I)G_D \geq 0, \quad 1 \leq D \leq 23. \quad (28)$$

Appendix A specifies and verifies it entirely by rational arithmetic.

## 5 A relative error bound for finite spheres

The limiting Gram matrices have kernels, so their positivity cannot be transferred by assuming that their ranges vary continuously. Instead, we compare the annihilators themselves. An exact diagonal change of variables reduces every error to a fixed finite family of pair-generated annihilators. This gives an error relative to  $H_Q$ , uniformly on the full Fock space.

Geometry-dependent diagonal changes of occupation-number coordinates already relate pseudopotential annihilators and their zero modes [18, Section II.E]. For a gap estimate, the needed additional statement is quantitative: on a fixed set of modes the change of variables approaches the identity, and its error is controlled by pair energy. The proof below establishes that statement before adding spectator modes and averaging over rotations.

For the local argument, let  $\mathcal{F}_{\text{loc}}$  be the exterior algebra of  $\mathbb{C}^{25}$  with orthonormal basis  $e_0, \dots, e_{24}$ . For  $Q \geq 24$  we identify it with the Fock space of the first 25 orbital modes in  $U_Q$ . By (9), each limiting pair vector  $v_p^*$  with  $p \leq 23$  lies in  $\wedge^2 \mathbb{C}^{25}$ . Contraction by these vectors is defined on  $\mathcal{F}_{\text{loc}}$  by the same exterior-algebra convention as before.

**Lemma 5.1** (Compression and annihilator bounds). *Let  $W$  be a linear map between finite-dimensional Hilbert spaces,  $G = W^\dagger W$ , and let  $C$  be self-adjoint on the domain of  $W$ . Then*

$$GCG \geq 0 \iff WCW^\dagger \geq 0.$$

On  $\mathcal{F}_{\text{loc}}$ , for any fixed finite collection  $A_b = c_k c_j B(v_p^*)$  with  $0 \leq p \leq 23$  and  $0 \leq j, k \leq 24$ ,

$$\sum_b \|A_b \psi\|^2 \leq C_0 \langle \psi, h_* \psi \rangle, \quad h_* := \sum_{p=0}^{23} B(v_p^*)^\dagger B(v_p^*), \quad (29)$$

where  $C_0$  depends only on the collection.

*Proof.* The two positivity statements both assert that the quadratic form of  $C$  is nonnegative on  $\text{ran } G = \text{ran } W^\dagger$ . For the second assertion, use  $\|c_k c_j B(v_p^*) \psi\| \leq \|B(v_p^*) \psi\|$  and sum; one may take  $C_0$  to be the largest multiplicity of a pair label  $p$ .  $\square$

**Lemma 5.2** (Uniform transfer). *For  $Q \geq 24$  and  $1 \leq D \leq 23$ , put*

$$C_D^Q = (d_D/d_a) \chi - E_D^Q + \varepsilon I, \quad h_Q = \sum_{p=0}^{23} B_p^\dagger B_p.$$

Assuming (28), there is a scalar  $\rho_Q \geq 0$ , independent of  $D$  and tending to zero as  $Q \rightarrow \infty$ , such that

$$\sum_{r,s \in \mathcal{R}_D} (C_D^Q)_{rs} B(w_{Dr}^Q)^\dagger B(w_{Ds}^Q) \geq -\rho_Q h_Q \quad (30)$$

on the entire Fock space  $\mathcal{F}_Q$ .

*Proof.* An exact change of variables. Take  $Q \geq 24$ . All annihilators in (30), including those in  $h_Q$ , involve only the modes  $0, \dots, 24$ . First work on  $\mathcal{F}_{\text{loc}}$ . Define the positive diagonal operator

$$\Delta_Q |A\rangle = \left( \prod_{x \in A} d_Q(x) \right) |A\rangle, \quad d_Q(x) = \sqrt{(Q)_x / Q^x},$$

on each basis wedge  $|A\rangle$ . This local Fock space is fixed as  $Q$  varies. Thus  $\Delta_Q$  and  $\Delta_Q^{-1}$  converge in norm to  $I$ ; also  $0 < \Delta_Q \leq I$ .

The pair coefficient formula (8) gives

$$v_p(x, y) = r_p(Q) d_Q(x) d_Q(y) v_p^*(x, y), \quad r_p(Q) = \sqrt{(2Q)^p / (2Q-2)_p} \geq 1.$$

Checking the factors attached to the removed orbital labels on a basis wedge gives the exact identities

$$B_p = r_p(Q) \Delta_Q^{-1} B(v_p^*) \Delta_Q, \quad (31)$$

$$B(v_p \wedge e_j \wedge e_k) = \frac{r_p(Q)}{d_Q(j) d_Q(k)} \Delta_Q^{-1} B(v_p^* \wedge e_j \wedge e_k) \Delta_Q. \quad (32)$$

These are identities on the local Fock space, with the inverse diagonal acting on the output particle sector. Repeated removed labels make both sides zero. In particular, solving (31) for  $B(v_p^*) \Delta_Q$  and using  $r_p(Q) \geq 1$  and  $\|\Delta_Q\| \leq 1$  gives

$$\Delta_Q h_* \Delta_Q \leq h_Q. \quad (33)$$

*The error is controlled by pair energy.* Fix  $D$ . Index a finite family by  $b = (p, j, k)$  with  $p + j + k = D$  and  $j < k$ , and put

$$A_b = B(v_p^* \wedge e_j \wedge e_k), \quad \beta_{rb}^Q = S_r^Q(D, D; p, j, k) \frac{r_p(Q)}{d_Q(j) d_Q(k)}.$$

Equations (26) and (32) yield

$$B(w_{Dr}^Q) = \Delta_Q^{-1} \left( \sum_b \beta_{rb}^Q A_b \right) \Delta_Q.$$

The coefficients are real, and Lemma 3.3 gives  $\beta_{rb}^Q \rightarrow \beta_{rb}^* := S_r(D, D; p, j, k)$ . Hence the finite real symmetric matrices

$$M_Q = (\beta^Q)^T C_D^Q \beta^Q \quad \text{converge to} \quad M_* = (\beta^*)^T (2\chi - E_D + \varepsilon I) \beta^*.$$

The quadratic expression in (30) is consequently

$$\Delta_Q \left( \sum_{b,c} (M_Q)_{bc} A_b^\dagger \Delta_Q^{-2} A_c \right) \Delta_Q.$$

Its limiting expression before the outer diagonals is

$$\sum_{b,c} (M_*)_{bc} A_b^\dagger A_c = \sum_{r,s} (2\chi - E_D + \varepsilon I)_{rs} B(w_{Dr}^*)^\dagger B(w_{Ds}^*) \geq 0. \quad (34)$$

Indeed, the certificate (28) and Lemma 5.1, applied to the map with columns  $w_{Dr}^*$ , give positivity before lifting; the lift  $\mathcal{L}_4$  preserves positivity.

For clarity, the perturbation estimate does not require  $M_* \geq 0$ . Let  $R_Q$  be the block matrix with entries

$$(R_Q)_{bc} = (M_Q)_{bc} \Delta_Q^{-2} - (M_*)_{bc} I.$$

Regard  $R_Q$  as an operator on the Hilbert direct sum of one copy of  $\mathcal{F}_{\text{loc}}$  for each index  $b$ . The diagonal  $\Delta_Q^{-2}$  acts on the output of  $A_c$ , hence on the sector with four fewer particles; using its direct sum over local sectors makes this a single operator identity. Its norm tends to zero, since both its index set and the local Fock space are fixed. For every vector  $\phi$ ,

$$\left| \sum_{b,c} \langle A_b \phi, (R_Q)_{bc} A_c \phi \rangle \right| \leq \|R_Q\| \sum_b \|A_b \phi\|^2 \leq C_0 \|R_Q\| \langle \phi, h_* \phi \rangle.$$

Combine this estimate with (34), take  $\phi = \Delta_Q \psi$ , and apply (33). This proves (30) on the local Fock space with a coefficient tending to zero. Taking a maximum over the fixed set  $1 \leq D \leq 23$  gives a common  $\rho_Q$ .

*Additional particles are spectators.* Order the 25 local modes before all other modes in the Fock factorization. Under the isometry that sends a local wedge tensored with a spectator wedge to their ordered product, each  $c_j$  with  $j \leq 24$  is its local annihilator tensored with the identity. The fermionic parity factor belongs to annihilators in the later spectator modes, which do not occur here. Thus every operator in (30) acts on the local factor alone. Tensoring the local inequality with the identity proves it on  $\mathcal{F}_Q$ , with the same  $\rho_Q$ , regardless of the number of occupied modes outside this factor.  $\square$

The allowance  $\varepsilon I$  in the certificate must also be paid for in pair energy. Here it is useful to sum over the orthonormal highest vectors *before* estimating their wedges. Bessel's inequality then avoids an extra factor equal to the number of spin copies.

**Proposition 5.3** (Four-body estimate). *For the coefficients in Appendix A,*

$$A_4 \leq \mathcal{L}_4(W_4 W_4^\dagger) + (1222\varepsilon + o(1))H, \quad Q \rightarrow \infty, \quad (35)$$

*uniformly on  $\mathcal{F}_Q$ .*

*Proof.* Haar averaging the matrix unit  $|D, D; r\rangle_Q \langle D, D; s|_Q$  gives the corresponding copy matrix unit tensored with the spin identity, divided by  $d_D$ . Therefore, on the retained spin blocks, the target minus  $A_4$ , with the  $\varepsilon I$  allowance added, is

$$d_a \sum_{D=1}^{23} \int \sum_{r,s} (C_D^Q)_{rs} B(w_{Dr}^Q(g))^\dagger B(w_{Ds}^Q(g)) dg.$$

Lemma 5.2, conjugated by each rotation, bounds this below by  $-o(1)H$ , because Schur averaging gives

$$d_a \int h_Q(g) dg = 24H.$$

The number of retained blocks is fixed, so this error remains uniform in particle number.

To bound the allowance, fix  $D$  and write

$$|D, D; r\rangle_Q = \sum_{\substack{p+j+k=D \\ j < k}} S_{rb}^Q v_p \otimes (e_j \wedge e_k), \quad b = (p, j, k).$$

The uncoupled vectors on the right are orthonormal. So are the highest vectors on the left: distinct  $r$  belong to orthogonal spin summands of the second pair factor. Thus  $S^Q (S^Q)^T = I$ . Apply the contraction  $S^Q \otimes I$  to the Hilbert-space-valued vector  $(c_k c_j B_p \psi)_b$ . It follows that

$$\sum_{r \in \mathcal{R}_D} \|B(w_{Dr}^Q) \psi\|^2 \leq \sum_{\substack{p+j+k=D \\ j < k}} \|c_k c_j B_p \psi\|^2 \leq \sum_{\substack{p+j+k=D \\ j < k}} \|B_p \psi\|^2.$$

This uses orthogonality before wedging; the vectors  $w_{Dr}^Q$  themselves need not be orthogonal. The number of triples on the right is

$$n_D = \sum_{n=1}^D \left\lceil \frac{n}{2} \right\rceil = \left\lfloor \frac{(D+1)^2}{4} \right\rfloor.$$

Conjugate by rotations and integrate. Each pair norm contributes  $\langle \psi, H \psi \rangle / d_a$ , so the total allowance is bounded by

$$\varepsilon \sum_{D=1}^{23} n_D H = \varepsilon \left( \sum_{m=1}^{12} m^2 + \sum_{m=1}^{11} m(m+1) \right) H = 1222\varepsilon H.$$

The discarded target spin blocks are positive. Removing the allowance therefore proves (35).  $\square$

## 6 The tail estimate and the gap

The finite comparisons are now in place. We first bound all remaining three-body blocks by a summable multiple of  $H_Q$ , then identify the zero mode at Laughlin flux.

**Lemma 6.1** (Three-body tail). *For sufficiently large  $Q$ ,*

$$\mathcal{L}_3 \left( W_3 \sum_{z=16}^Q q_z(Q) R_{3,z} W_3^\dagger \right) \geq -\delta_Q H, \quad (36)$$

where

$$\delta_Q = \sum_{\substack{17 \leq z \leq Q \\ z \text{ odd}}} \frac{d_{3,z}}{d_a} |q_z(Q)| (z+1), \quad \delta_Q \longrightarrow \frac{61}{4096}. \quad (37)$$

*Proof.* For  $16 \leq z \leq Q$ , the bracket in (10) is positive, since it is at least  $2z - 2$ . Thus only odd  $z$  contribute negatively. A normalized highest vector of  $R_{3,z}$  is  $\sum_{p=0}^z s_p v_p \otimes e_{z-p}$ . Its orbit resolves that spin projector with factor  $d_{3,z}$ . After wedging and lifting, its quadratic form is

$$\left\| \sum_{p=0}^z s_p c_{z-p}(g) B_p(g) \psi \right\|^2 \leq \sum_{p=0}^z \|B_p(g) \psi\|^2.$$

Averaging each summand proves (36) with (37).

For  $Q \geq 4$ , the ratios in  $(Q)_z / (2Q - 2)_z$  are nonincreasing, so

$$\frac{(Q)_z}{(2Q - 2)_z} \leq \left( \frac{Q}{2Q - 2} \right)^z \leq (2/3)^z, \quad \frac{d_{3,z}}{d_a} \leq 2.$$

The summands, extended by zero for  $z > Q$ , are therefore dominated by  $6z(z+1)(2/3)^z$ . Dominated convergence and (11) give

$$\lim_{Q \rightarrow \infty} \delta_Q = \frac{3}{2} \sum_{z=17,19,\dots} (3z-1)(z+1)2^{-z} = \frac{61}{4096}.$$

For the last identity, write  $z = 17 + 2m$ . Then  $(3z-1)(z+1) = 900 + 208m + 12m^2$ , while the sums of  $4^{-m}$ ,  $m4^{-m}$ , and  $m^2 4^{-m}$  are  $4/3$ ,  $4/9$ , and  $20/27$ , respectively. Multiplication by  $(3/2)2^{-17}$  gives  $61/4096$ .  $\square$

*Proof of Theorem 1.1.* Combine Lemma 2.1, (18), (22), Proposition 5.3, and Lemma 6.1. They give

$$H^2 \geq d_a \int K(g) dg + (1 - \eta - \delta_Q - 1222\varepsilon - o(1))H.$$

The integral is positive. The certificate gives

$$\eta = \frac{93527408868499}{10^{14}}, \quad \varepsilon = \frac{3}{10^6},$$

so the coefficient tends to

$$1 - \frac{93527408868499}{10^{14}} - \frac{61}{4096} - 1222 \frac{3}{10^6} = \gamma_* = \frac{4616733319001}{10^{14}} > \frac{1}{25}. \quad (38)$$

The limit in (38) is  $\gamma_*$ . For each fixed  $0 < \gamma < \gamma_*$ , the coefficient therefore exceeds  $\gamma$  for all sufficiently large  $Q$ . Every convergence threshold is independent of particle number, proving the assertion on  $\mathcal{F}_Q$ .  $\square$

We now recover the known unique zero mode of Haldane's spherical parent Hamiltonian [7, p. 606, Equations (6)–(7)]. The elementary divisibility proof also fixes the exact flux at which the kernel is a line.

**Lemma 6.2** (Kernel at Laughlin flux). *For  $N \geq 2$  and  $Q = 3(N - 1)$ , the kernel of  $H_{N,Q}$  is the line spanned by  $\Psi_{L,N}$ .*

*Proof.* Realize  $U_Q$  as the homogeneous polynomials of degree  $Q$  in spinor coordinates  $(u, v)$ . Write  $b_{ij} = u_i v_j - u_j v_i$ . The pair-spin  $Q - r$  summand has highest polynomial

$$b_{ij}^r u_i^{Q-r} u_j^{Q-r}.$$

Since  $b_{ij}$  is invariant under simultaneous rotations of the pair, every vector in this summand is divisible by  $b_{ij}^r$ . Exchange of the pair has parity  $(-1)^r$ , so antisymmetric pair states have odd  $r$ .

If  $H_{N,Q}\Psi = 0$ , then, in the ambient tensor space,

$$0 = \langle \Psi, H_{N,Q}\Psi \rangle = \sum_{i < j} \|P_{ij}^{(1)}\Psi\|^2.$$

Thus  $P_{ij}^{(1)}\Psi = 0$  for every pair. Expanding in the other particles' monomials, the remaining pair sectors all have odd  $r \geq 3$ . Thus  $b_{ij}^3$  divides  $\Psi$  for every  $i < j$ . Each bracket  $b_{ij} = v_j u_i - u_j v_i$  is primitive as a polynomial in  $u_i$ , since  $v_j$  and  $u_j v_i$  are coprime, and is linear over the fraction field of the other variables. Gauss's lemma therefore makes it irreducible in the polynomial ring. Brackets for distinct unordered pairs are nonassociate. They are consequently relatively prime. Unique factorization implies

$$\prod_{i < j} b_{ij}^3 \mid \Psi.$$

This product has degree  $3(N - 1) = Q$  in each particle's spinor variables, exhausting the multidegree of  $\Psi$ ; the quotient is constant.

Conversely,  $\Psi_{L,N}$  is antisymmetric. Its factor  $b_{ij}^3$  leaves pair degrees  $Q - 3$  in each spinor, whose maximum total pair spin is  $Q - 3$ . Multiplication by the invariant bracket cannot introduce spin  $Q - 1$ . Hence every pair projector annihilates  $\Psi_{L,N}$ .  $\square$

*Proof of Corollary 1.2.* Apply Theorem 1.1 with  $\gamma = 1/25$ . On each particle sector, the spectral theorem gives

$$H_{N,Q} \geq \frac{1}{25}(I - P_{\ker H_{N,Q}})$$

once  $Q$  exceeds the common threshold. Take  $Q = 3(N - 1)$  and apply Lemma 6.2. Choosing  $N_0$  so that  $3(N_0 - 1)$  exceeds the threshold proves the claim.  $\square$

The following comparison specializes the recursive spectral relations of Lemm, Nachtergaele, Warzel, and Young [13, arXiv v1, Theorem 2.1(I), Lemma 2.3, and (2.10)].

**Corollary 6.3** (Charge versus neutral gap at fixed flux). *For  $N \geq 2$ , fix  $Q_N = 3(N - 1)$  and write  $E_0(n; Q_N) = \min \text{spec } H_{n,Q_N}$  for the coefficient-one interaction (1). Let  $\text{Gap}_N$  be the first positive eigenvalue of  $H_{N,Q_N}$ , its neutral gap. The particle-number charge gap at this same flux,*

$$\Delta_N = E_0(N + 1; Q_N) + E_0(N - 1; Q_N) - 2E_0(N; Q_N),$$

*satisfies*

$$\Delta_N = E_0(N + 1; Q_N) \geq \frac{N}{N - 1} \text{Gap}_N > 0.$$

*In particular,  $\Delta_N \geq N/[25(N - 1)]$  for  $N \geq N_0$ , with  $N_0$  as in Corollary 1.2.*

*Proof.* All three sectors are nonempty since  $N + 1 \leq \dim U_{Q_N} = 3N - 2$ . Lemma 6.2 gives  $E_0(N; Q_N) = 0$  and a one-dimensional ground space. The  $N$ -particle sector has dimension greater than one, so  $\text{Gap}_N > 0$ . For a nonzero Laughlin vector  $\Psi$ , each pair annihilator  $B_p$  commutes with every  $c_j$ , while  $\sum_j \|c_j \Psi\|^2 = N \|\Psi\|^2$ . Thus some  $c_j \Psi$  is a nonzero  $(N - 1)$ -particle zero mode, giving  $E_0(N - 1; Q_N) = 0$ . The divisibility argument in Lemma 6.2 excludes a zero mode in the  $(N + 1)$ -particle sector: the required product of cubed brackets would have degree  $3N > Q_N$  in each particle variable.

Let  $P_n$  project onto the lowest-energy eigenspace of  $H_{n, Q_N}$ , and define the overlap operator on the  $(N + 1)$ -particle sector by

$$\mathcal{G}_N = P_{N+1} \left( \sum_{j=0}^{Q_N} c_j^\dagger P_N c_j \right) P_{N+1}.$$

Here  $P_{N+1}$  projects onto a positive-energy ground eigenspace. For the pure two-body fermionic form (6), the cited Theorem 2.1(I), with  $m_0 = m_1 = 2$ , gives

$$E_0(N + 1; Q_N) \geq \frac{N + 1}{N - 1} E_0(N; Q_N) + \frac{N + 1 - \|\mathcal{G}_N\|}{N - 1} \text{Gap}_N.$$

The cited Lemma 2.3 gives  $\|\mathcal{G}_N\| \leq \text{rank } P_N = 1$ . Since  $E_0(N; Q_N) = 0$ , this yields  $E_0(N + 1; Q_N) \geq N \text{Gap}_N / (N - 1)$ . The two zero energies give the asserted second difference, and Corollary 1.2 supplies the final uniform bound.  $\square$

## 7 A planar corollary

The full-Fock-space estimate also gives a gap on the plane. We use the analytic lowest-Landau-level representation described by Girvin and Jach [5, Section II], with Gaussian scale fixed below. Let  $\mathcal{B}$  be the Bargmann space of entire functions with norm

$$\|f\|_{\mathcal{B}}^2 = \frac{1}{\pi} \int_{\mathbb{C}} |f(z)|^2 e^{-|z|^2} d^2 z, \quad e_j(z) = \frac{z^j}{\sqrt{j!}}.$$

Multiplication by  $\pi^{-1/2} e^{-|z|^2/2}$  identifies this space unitarily with the planar lowest Landau level at magnetic field 2; changing magnetic length is a unitary dilation. For a pair of coordinates set  $w = (z_1 - z_2)/\sqrt{2}$  and  $Z = (z_1 + z_2)/\sqrt{2}$ . Let  $P^{(1)}$  be the orthogonal projector onto relative degree one, with no restriction on the degree in  $Z$ , and put

$$H_N^{\text{pl}} = \sum_{1 \leq i < j \leq N} P_{ij}^{(1)} \Big|_{\bigwedge^N \mathcal{B}}.$$

Thus each pair projector has coefficient one, as on the sphere.

**Corollary 7.1** (Planar spectral gap). *For every  $N \geq 2$ , the untruncated planar Hamiltonian satisfies*

$$(H_N^{\text{pl}})^2 \geq \gamma_* H_N^{\text{pl}},$$

where  $\gamma_*$  is the constant in Theorem 1.1. Equivalently, if  $P_{0,N}^{\text{pl}}$  projects onto its kernel, then  $H_N^{\text{pl}} \geq \gamma_*(I - P_{0,N}^{\text{pl}})$ . The constant is independent of both particle number and total angular momentum. In particular, the case  $\ell = 3$  of the spectral gap conjecture in [21, Conjecture A.1] holds, in its pair-projector normalization.

*Proof.* The coordinate change  $(z_1, z_2) \mapsto (w, Z)$  is unitary for the Bargmann inner product. Consequently the vectors  $wZ^p/\sqrt{p!}$ ,  $p \geq 0$ , form an orthonormal basis of the relative-degree-one subspace. Their coefficients in the normalized wedge basis are

$$v_p^{\text{pl}}(i, j) = (i - j) \sqrt{\frac{p!}{2^p i! j!}} \mathbf{1}_{i+j=p+1}, \quad i < j.$$

Indeed, expanding  $(z_1 - z_2)(z_1 + z_2)^p$  first gives the coefficient in the tensor basis; passage to the normalized antisymmetric wedge multiplies it by  $\sqrt{2}$ . These are exactly the limits in (9). The second-quantized pair-projector identity therefore gives  $H_N^{\text{pl}} = \sum_{p \geq 0} B(v_p^{\text{pl}})^\dagger B(v_p^{\text{pl}})$  on antisymmetric polynomials.

Fix  $N$  and a total degree  $L$ . The corresponding homogeneous subspace has the finite orthonormal basis

$$e_{i_1} \wedge \cdots \wedge e_{i_N}, \quad 0 \leq i_1 < \cdots < i_N, \quad \sum_{k=1}^N i_k = L.$$

For  $Q \geq \max\{L, N - 1, 2\}$  this is also the entire spherical subspace of total lowering deficit  $L$ , under the identification of normalized one-particle bases. Both Hamiltonians preserve this subspace: a pair is annihilated and created with the same sum of orbital labels. Only  $p + 1 \leq L$  can contribute. The finitely many matrix entries of the spherical restriction therefore converge to those of the planar restriction by (8)–(9); the convergence is in operator norm.

Fix  $0 < \gamma < \gamma_*$ . Theorem 1.1 gives  $H_Q^2 \geq \gamma H_Q$  for all sufficiently large  $Q$ . Restrict to the invariant block just described and let  $Q \rightarrow \infty$ . Norm convergence passes through the square and gives  $(H_N^{\text{pl}})^2 \geq \gamma H_N^{\text{pl}}$  on that block. Now let  $\gamma \uparrow \gamma_*$ ; the cone of positive matrices is closed, so the same inequality holds at  $\gamma_*$ . The limit  $Q \rightarrow \infty$  is taken separately for each fixed  $\gamma$ , allowing the flux threshold to depend on  $\gamma$ . Here  $N$  and  $L$  were arbitrary, and the resulting constant is the same for all of them.

Finally,  $\bigwedge^N \mathcal{B}$  is the Hilbert direct sum of its homogeneous subspaces, and  $0 \leq H_N^{\text{pl}} \leq \binom{N}{2} I$ . The block inequalities thus extend by continuity to the whole space. The spectral theorem gives the asserted gap. Restriction to total degrees  $L \leq 3N(N - 1)/2$  gives exactly the sector in the cited conjecture for  $\ell = 3$ .  $\square$

## A The finite rational certificate

This appendix specifies the seven rows and all arithmetic needed to verify (21) and (28). Starting from the integer rows in Section A.1, evaluate the rational three-body traces in Section A.2 and the rational four-body matrices in Section A.3. The margin and coefficient-sign tables in those sections record the resulting checks. The formulas below determine every value using only integers and rational numbers.

### A.1 Row data

Let  $P = 10^7$ . The first two numbers of each row are  $t$  and  $\ell = P\lambda$ . An entry  $pj : a$  specifies the integer  $a_{pj}$ , with  $p$  and  $j$  parsed as separate single-digit labels; omitted entries are zero. Continuation lines belong to the preceding row. Define

$$\alpha_{pj} = \frac{a_{pj}}{P} \sqrt{\frac{2^p \binom{p+j}{p}}{2^t \binom{p+j}{t}}} = \frac{a_{pj}}{P} \sqrt{\frac{2^{p-t} t! (p+j-t)!}{p! j!}}. \quad (39)$$

Every nonzero entry has  $0 \leq p + j - t \leq 8$ .

```

0 7181518 05:-849651 07:-2958744 15:751659 16:-43084 17:-146216 32:720949 35:-127514
  42:222437 43:-262822 53:-117700 61:-55024 71:-166049
1 -112991 05:142772 06:-142342 07:527055 08:50742 16:197917 18:-18343 24:5764 27:-21183
  32:75832 35:-6115 45:2660 53:-9455 61:-27091 63:-3494 71:479 72:-15331
1 -4247311 05:-12207250 06:-14064191 07:-14063663 08:-15662679 16:483831 18:3987214
  24:-814558 27:888887 32:966462 35:145978 45:-171722 53:34350 61:389319 63:97603
  71:412447 72:-53786
2 3438746 14:6689266 15:6620205 16:6625065 17:4969272 18:4878238 27:-1387380 36:-586188
  37:344087 42:-1120634 45:-318530 46:559169 55:157455 61:1207103 63:211617 64:33240
  71:577001 73:-142578
5 -98645 05:-2719776 08:-8519913 14:-949523 17:-1952105 18:-1562606 28:-8919966 32:248826
  35:-1017623 37:3656005 38:-2081871 42:-33163 43:-305905 46:35677 47:-1320749
  48:-1040378 53:502094 55:1060633 58:-998003 61:-113617 63:184919 64:209239 65:-68198
  67:-266297 72:-606696 73:-563620 74:437345 76:-432189
5 -3160276 05:-1698129 08:10575875 14:-252576 17:-8002616 18:22592844 28:7370640
  32:289294 35:1400647 37:1987740 38:4030699 42:-1731401 43:-1034062 46:-1106371
  47:-1257051 48:1461735 53:1408149 55:1136847 58:1878501 61:-2838516 63:-734099
  64:1062416 65:11745 67:611312 72:-480776 73:20459 74:316925 76:37092
7 1441784 07:3441286 08:-6097155 37:-1002399 47:-251321 58:1143225 66:-17397 71:-1007464
  73:-662286 76:-344621 78:-1115284

```

In particular  $P^2\eta = \sum_{\rho} \ell_{\rho}^2 = 93527408868499$ .

## A.2 Three-body arithmetic

For  $c \in \{1, 2\}$ , an integer  $T \geq 0$ , and integers  $z, p$ , define

$$\mathbf{U}(c, z, T, p) = \sum_{h=\max(0, p+z-T)}^{\min(p, z)} (-1)^{z-h} \binom{z}{h} \binom{T-z}{p-h} c^{p-h}, \quad (40)$$

and set it to zero if  $z$  or  $p$  lies outside  $[0, T]$ . Expanding (13) gives

$$s_{z, T, p}^{(1/\sqrt{3})} = \mathbf{U}(2, z, T, p) \sqrt{\frac{2^z \binom{T}{z}}{3^T 2^p \binom{T}{p}}}.$$

Substitution in (20) and (39) cancels all radicals:

$$P^2 e_z = \sum_{\rho} \sum_{T=\max(z, t_{\rho})}^{15} \frac{2^z \binom{T}{z}}{3^T 2^{t_{\rho}} \binom{T}{t_{\rho}}} X_{\rho, T} (X_{\rho, T} + 2\ell_{\rho} \mathbf{U}(2, z, T, t_{\rho})), \quad (41)$$

where  $X_{\rho, T} = \sum_{p+j=T} a_{pj}^{\rho} \mathbf{U}(2, z, T, p)$ , with value zero on empty levels. For each  $z \in \{2, 4, 5, \dots, 15\}$ , evaluate (41) and form the exact rational margin  $m_z = \frac{3}{2}(3z-1)(-1/2)^z - e_z$ . Write  $10^6 m_z = n_z/d_z$  with integers  $n_z, d_z$  and  $d_z > 0$ . Integer division gives the floors  $\lfloor n_z/d_z \rfloor$  listed below:

| $z$   | 2       | 4      | 5   | 6   | 7   | 8   | 9   | 10  | 11  | 12  | 13  | 14   | 15  |
|-------|---------|--------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|-----|
| floor | 1298313 | 211317 | 249 | 250 | 249 | 250 | 250 | 250 | 249 | 250 | 250 | 6470 | 250 |

All are positive, proving (21).

## A.3 Four-body arithmetic

Fix  $1 \leq D \leq 23$ ; all copy indices below lie in  $\mathcal{R}_D$ . Put

$$u_r = \frac{1}{r!(D-r)!}, \quad (E_D)_{rs} = \sqrt{u_r u_s} Y_{rs}, \quad (G_D)_{rs} = \sqrt{u_r u_s} Z_{rs}.$$

The matrices  $Y, Z$  are rational. Define

$$\mathbf{V}_r(D, T; p, j, k) = \mathbf{U}(1, r, j+k, j) \mathbf{U}(1, D-r, T-r, p),$$

with value zero for  $r > j+k$ . Two applications of (13) give

$$S_r(D, T; p, j, k) = \mathbf{V}_r(D, T; p, j, k) \sqrt{2^{1-(j+k)-T+r} \frac{j!k!p!u_r}{(T-D)!}}. \quad (42)$$

Therefore

$$\begin{aligned} P^2 Y_{rs} = & - \sum_{\rho} \sum_{\substack{p,j;q,l \\ T \geq D}} a_{pj}^{\rho} a_{ql}^{\rho} \mathbf{V}_r(D, T; p, j, k) \mathbf{V}_s(D, T; q, l, i) \\ & \times 2^{1-2T+p+q-t_{\rho}+(r+s)/2} \frac{i!k!t_{\rho}!}{(T-D)!}, \end{aligned} \quad (43)$$

where  $i = p + j - t_{\rho}$ ,  $k = q + l - t_{\rho}$ , and  $T = p + j + k$ . Both entry sums are ordered. To verify the cancellation, multiply the two factors in (42) by the two factors (39); the factorial product under the combined square root is  $(t_{\rho}!i!k!)^2 u_r u_s / ((T-D)!)^2$ . Using  $j+k = T-p$  and  $l+i = T-q$ , its power of two has exponent  $2 + 2p + 2q - 2t_{\rho} - 4T + r + s$ . Taking the positive square root and removing  $\sqrt{u_r u_s}$  gives exactly the displayed factor.

For an increasing quadruple  $A$  of nonnegative indices with sum  $D+1$ , put  $f(A) = \prod_{b \in A} b!$ . If  $(x, y, j, k)$  lists its elements, let  $\sigma(x, y, j, k)$  be the sign of sorting this list into increasing order. In each summand below put  $p = x + y - 1$ , and define

$$L_r(A) = 2^{(1-2D+r)/2} \sum_{\substack{x < y \text{ in } A \\ \{j,k\} = A \setminus \{x,y\}, j < k}} \sigma(x, y, j, k) (x-y) p! j! k! \mathbf{V}_r(D, D; p, j, k). \quad (44)$$

Combining (9) and (42), the coefficient of  $w_{D_r}^*$  on the Slater wedge  $A$  is  $\sqrt{u_r/f(A)} L_r(A)$ . Thus

$$Z_{rs} = \sum_{\substack{A \text{ increasing quadruple} \\ \sum_{b \in A} b = D+1}} \frac{L_r(A) L_s(A)}{f(A)}. \quad (45)$$

All exponents of two in these formulas are integers, since  $r, s$  are odd.

Let  $D_u = \text{diag}(u_r)$  and  $D_{\sqrt{u}} = \text{diag}(\sqrt{u_r})$ . The matrix in (28) equals  $D_{\sqrt{u}} M D_{\sqrt{u}}$ , where

$$M = Z B Z, \quad B = \text{diag}((2\mathbf{1}_{r=1} + \varepsilon) u_r) - D_u Y D_u. \quad (46)$$

Thus it suffices to check  $M \geq 0$  using rational arithmetic. This congruence uses only the positive diagonal  $D_{\sqrt{u}}$ ; it does not invert  $G_D$  or presume that its columns are independent.

To check positivity for each  $D = 1, \dots, 23$ , form  $Y$  and  $Z$  from (43) and (45), then form  $B$  and  $M$  from (46), using  $\varepsilon = 3/10^6$ . All sums range over the printed rows and the finite index sets specified above. Compute with exact fractions throughout. If  $d = |\mathcal{R}_D|$ , initialize  $J = I_d$ . For  $b = 1, \dots, d$ , compute in order

$$X = M J, \quad c_b = \frac{\text{tr } X}{b}, \quad J = c_b I_d - X. \quad (47)$$

This is the Leverrier–Faddeev recurrence applied to  $-M$ ; see [9]. It gives  $\det(xI + M) = x^d + c_1 x^{d-1} + \dots + c_d$ . For example, this recurrence follows by multiplying the polynomial adjugate of  $xI + M$  by  $xI + M$  and comparing coefficients, together with the derivative-of-determinant trace identity. Exact evaluation of (40), (43), (44), (45), and (47) gives:

| $D$ | signs | $D$ | signs    | $D$ | signs       |
|-----|-------|-----|----------|-----|-------------|
| 1   | 0     | 9   | ++000    | 17  | ++++++000   |
| 2   | 0     | 10  | +0000    | 18  | +++++0000   |
| 3   | 00    | 11  | +++000   | 19  | ++++++000   |
| 4   | 00    | 12  | ++0000   | 20  | ++++++0000  |
| 5   | +00   | 13  | ++++000  | 21  | +++++++000  |
| 6   | 000   | 14  | +++0000  | 22  | +++++++0000 |
| 7   | +000  | 15  | +++++000 | 23  | +++++++000  |
| 8   | +000  | 16  | ++++0000 |     |             |

The signs list the coefficients after the leading 1, in descending degree. For each computed coefficient, choose a positive denominator and inspect its integer numerator: + denotes a positive numerator and 0 denotes zero. We use the characteristic-coefficient positivity test for symmetric matrices [19, Theorem 5]. Every coefficient is nonnegative, so  $\det(xI + M) > 0$  for  $x > 0$ . Since  $M$  is real symmetric, a negative eigenvalue would give a positive root of this polynomial. Therefore  $M \geq 0$ , proving (28).

*Remark A.1* (Nature of the certificate). The data above certify fixed spin-matrix inequalities. The positive comparison operator is a sum of squares by construction. No many-particle finite-size spectrum, numerical extrapolation, or assumed plasma property enters the argument. The exact equalities and inequalities in this appendix can be verified by a finite calculation using only integers and rational numbers.

## References

- [1] Stephen D. Bartlett, Terry Rudolph, and Robert W. Spekkens. Reference frames, superselection rules, and quantum information. *Reviews of Modern Physics*, 79:555–609, 2007. doi:10.1103/RevModPhys.79.555.
- [2] Li Chen and Alexander Seidel. Algebraic approach to the study of zero modes of Haldane pseudopotentials. *Physical Review B*, 91:085103, 2015. doi:10.1103/PhysRevB.91.085103.
- [3] Esther Cruz, Flavio Baccari, Jordi Tura, Norbert Schuch, and J. Ignacio Cirac. Preparation and verification of tensor network states. *Physical Review Research*, 4:023161, 2022. doi:10.1103/PhysRevResearch.4.023161.
- [4] Qiang Gao, Ryan A. Lanzetta, Patrick Ledwith, Jie Wang, and Eslam Khalaf. Bootstrapping the quantum Hall problem. *Physical Review X*, 15:031034, 2025. doi:10.1103/csnn-vjhn.
- [5] S. M. Girvin and Terrence Jach. Formalism for the quantum Hall effect: Hilbert space of analytic functions. *Physical Review B*, 29(10):5617–5625, 1984. doi:10.1103/PhysRevB.29.5617.
- [6] S. M. Girvin, A. H. MacDonald, and P. M. Platzman. Magneto-roton theory of collective excitations in the fractional quantum Hall effect. *Physical Review B*, 33(4):2481–2494, 1986. doi:10.1103/PhysRevB.33.2481.
- [7] F. D. M. Haldane. Fractional quantization of the Hall effect: a hierarchy of incompressible quantum fluid states. *Physical Review Letters*, 51(7):605–608, 1983. doi:10.1103/PhysRevLett.51.605.

- [8] F. D. M. Haldane and E. H. Rezayi. Finite-size studies of the incompressible state of the fractionally quantized Hall effect and its excitations. *Physical Review Letters*, 54(3):237–240, January 1985. doi:10.1103/PhysRevLett.54.237.
- [9] Shui-Hung Hou. On the Leverrier–Faddeev algorithm. In *Electronic Proceedings of the Third Asian Technology Conference in Mathematics*, 1998. Full text.
- [10] Sabine Jansen. Fermionic and bosonic Laughlin state on thick cylinders. *Journal of Mathematical Physics*, 53:123306, 2012. doi:10.1063/1.4768250.
- [11] Stefan Knabe. Energy gaps and elementary excitations for certain VBS-quantum antiferromagnets. *Journal of Statistical Physics*, 52:627–638, 1988. doi:10.1007/BF01019721.
- [12] Robert B. Laughlin. Anomalous quantum Hall effect: an incompressible quantum fluid with fractionally charged excitations. *Physical Review Letters*, 50(18):1395–1398, 1983. doi:10.1103/PhysRevLett.50.1395.
- [13] Marius Lemm, Bruno Nachtergaele, Simone Warzel, and Amanda Young. Recursive spectral relations and the charge vs neutral gap in fractional quantum Hall systems. *Journal of Mathematical Physics*, 67(1):011901, January 2026. doi:10.1063/5.0292975.
- [14] Elliott H. Lieb, Nicolas Rougerie, and Jakob Yngvason. Rigidity of the Laughlin liquid. *Journal of Statistical Physics*, 172:544–554, 2018. doi:10.1007/s10955-018-2082-1.
- [15] Tahereh Mazaheri, Gerardo Ortiz, Zohar Nussinov, and Alexander Seidel. Zero modes, bosonization, and topological quantum order: The Laughlin state in second quantization. *Physical Review B*, 91:085115, 2015. doi:10.1103/PhysRevB.91.085115.
- [16] Bruno Nachtergaele, Simone Warzel, and Amanda Young. Spectral gaps and incompressibility in a  $\nu = 1/3$  fractional quantum Hall system. *Communications in Mathematical Physics*, 383:1093–1149, 2021. doi:10.1007/s00220-021-03997-0.
- [17] Masaaki Nakamura, Zheng-Yuan Wang, and Emil J. Bergholtz. Exactly solvable fermion chain describing a  $\nu = 1/3$  fractional quantum Hall state. *Physical Review Letters*, 109(1):016401, 2012. doi:10.1103/PhysRevLett.109.016401.
- [18] G. Ortiz, Z. Nussinov, J. Dukelsky, and A. Seidel. Repulsive interactions in quantum Hall systems as a pairing problem. *Physical Review B*, 88:165303, 2013. doi:10.1103/PhysRevB.88.165303.
- [19] Helfried Peyrl and Pablo A. Parrilo. Computing sum of squares decompositions with rational coefficients. *Theoretical Computer Science*, 409(2):269–281, 2008. doi:10.1016/j.tcs.2008.09.025.
- [20] Kshiti Sneh Rai, Ilya Kull, Patrick Emonts, Jordi Tura, Norbert Schuch, and Flavio Baccari. A hierarchy of spectral gap certificates for frustration-free spin systems. *Quantum*, 10:2065, 2026. doi:10.22331/q-2026-04-13-2065.
- [21] Nicolas Rougerie. On the Laughlin function and its perturbations. *Séminaire Laurent Schwartz — EDP et applications*, pages 1–17, 2018–2019. Exp. II. doi:10.5802/slsedp.131.
- [22] Severin Schraven and Simone Warzel. Fractional quantum Hall states: Infinite matrix product representation and its implications, 2026. arXiv:2601.12165v1, January 17, 2026.
- [23] S. A. Trugman and S. Kivelson. Exact results for the fractional quantum Hall effect with general interactions. *Physical Review B*, 31(8):5280–5284, 1985. doi:10.1103/PhysRevB.31.5280.

- [24] Zheng-Yuan Wang and Masaaki Nakamura. One-dimensional lattice model with an exact matrix-product ground state describing the Laughlin wave function. *Physical Review B*, 87:245119, 2013. [doi:10.1103/PhysRevB.87.245119](https://doi.org/10.1103/PhysRevB.87.245119).
- [25] Simone Warzel and Amanda Young. The spectral gap of a fractional quantum Hall system on a thin torus. *Journal of Mathematical Physics*, 63(4):041901, 2022. [doi:10.1063/5.0084677](https://doi.org/10.1063/5.0084677).
- [26] Simone Warzel and Amanda Young. A bulk spectral gap in the presence of edge states for a truncated pseudopotential. *Annales Henri Poincaré*, 24:133–178, 2023. [doi:10.1007/s00023-022-01210-z](https://doi.org/10.1007/s00023-022-01210-z).
- [27] G. C. Wick. The evaluation of the collision matrix. *Physical Review*, 80(2):268–272, 1950. [doi:10.1103/PhysRev.80.268](https://doi.org/10.1103/PhysRev.80.268).